



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2007

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:
(Please tick or highlight)

- ☐ Mr Ee
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	Total	Total
Mark	15	15	15	15	15	15	15	15	120	100

Mark

QUESTION 1 (15 marks)

- (a) Find $\int \frac{x}{\sqrt{16-x^2}} dx$ 2
- (b) By completing the square, find $\int \frac{8}{x^2+4x+13} dx$ 2
- (c) Use integration by parts to evaluate $\int_1^e x^4 \log_e x dx$. 4
- (d) Use the substitution $u = \cos x$ to find $\int \cos^2 x \sin^5 x dx$ 3
- (e) Express $\frac{3x+7}{(x+1)(x+2)(x+3)}$ in partial fractions and hence prove that $\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx = \ln 2$. 4

QUESTION 2 (15 marks) Start a new page

- (a) Let $z = 2+i$ and $w = 1-i$. Find, in the form $x+iy$,
- (i) $3z+iw$ 1
- (ii) $z\bar{w}$ 1
- (iii) $\frac{5}{z}$ 1
- (b) Let $\alpha = -\sqrt{3}+i$.
- (i) Express α in modulus-argument form. 2
- (ii) Express α^4 in modulus-argument form. 2
- (iii) Hence express α^4 in the form $x+iy$. 1

QUESTION 2 (Continued)

- (c) If $z_1 = 4 + i$ and $z_2 = 1 + 2i$ show geometrically how to construct the vectors representing.

(i) $z_1 + z_2$ 1

(ii) $z_1 - z_2$ 1

- (d) Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

(i) Find the coordinates of the foci and x-intercepts of the hyperbola. 2

(ii) Find the equations of the directrices and the asymptotes of the hyperbola. 2

(iii) What are the parametric equations of this hyperbola? 1

QUESTION 3 (15 marks) Start a new page

- (a) If α , β and γ are the roots of the cubic equation $x^3 + mx + n = 0$, find in terms of m and n , the values of

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 3

(ii) $\alpha^3 + \beta^3 + \gamma^3$ 2

(iii) Determine the cubic equation whose roots are α^2 , β^2 and γ^2 . 3

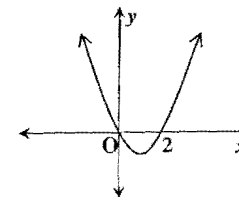
- (b) Given that the equation $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ has a triple root, find all the roots of the equation. 4

- (c) If $y = e^{-x}(A \sin 2x + B \cos 2x)$, prove that 3

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0$$

QUESTION 4 (15 marks) Start a new page

- (a) Given $f(x) = x^2 - 2x$. On separate diagrams sketch the graphs of the following. Indicate clearly any asymptotes, intercepts with the axes and local maxima and minima.



(i) $y = |f(x)|$ 1

(ii) $y = f(|x|)$ 1

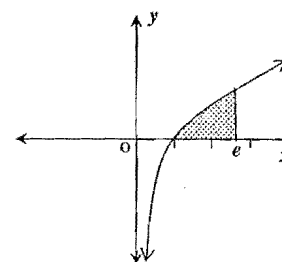
(iii) $y = \frac{1}{f(x)}$ 2

(iv) $y^2 = f(x)$ 2

(v) $y = [f(x)]^2$ 1

(vi) $y = \ln[f(x)]$ 2

- (b) The region bounded by $y = \ln x$, $x = e$ and the x -axis is rotated about the y -axis. Find the volume of rotation. Use the method of cylindrical shells. 3



- (b) First differentiating both sides of the formula

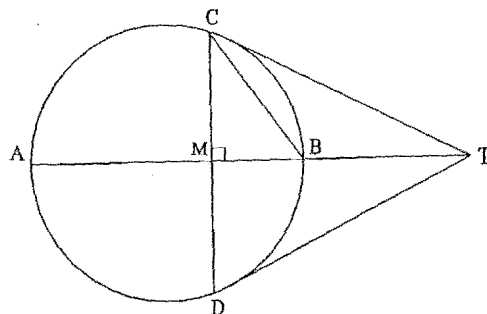
$$1 + x + x^2 + x^3 + \dots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

then find an expression for

$$1 + 2 \times 2 + 3 \times 4 + 4 \times 8 + \dots + n 2^{n-1}$$

QUESTION 5 (15 marks) Start a new page

- (a) In the circle shown below, the diameter AB meets the chord CD at right angles at M. The tangents at C and D meet at T.



(i) Show that BC bisects $\angle MCT$.

2

(ii) Show that triangle BCM is similar to triangle CAM. Hence, show that $CM^2 = AM \times BM$

2

(iii) Show that $TB \times TA = MB \times TA + TB \times TM$

4

(iv) Hence, or otherwise, show that M divides the interval AB internally in the same ratio that T divides AB externally.

2

(c) If $I_n = \int \tan^n x \, dx$

(i) Show that $I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$

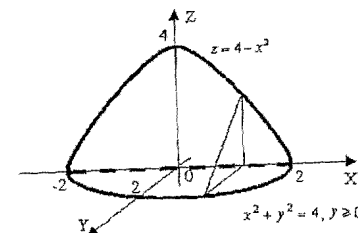
3

(ii) Find $\int \tan^6 x \, dx$.

2

QUESTION 6 (15 marks) Start a new page

- (a) The solid shown has a semicircular base of radius 2 units. Vertical cross-sections perpendicular to the diameter are right-angled triangles whose height is bounded by the parabola $z = 4 - x^2$.

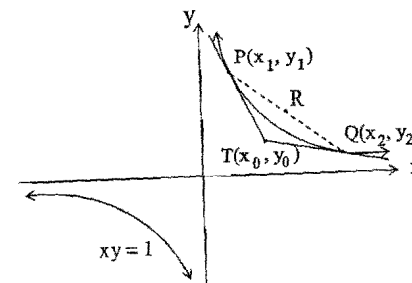


By slicing at right angles to the x -axis, show that the volume of the solid is given by

$$V = \int_0^2 (4 - x^2)^{3/2} dx, \text{ and hence calculate this volume.}$$

5

- (b) The tangents at $P(x_1, y_1)$, $Q(x_2, y_2)$ on the hyperbola $xy = 1$ intersect at the point $T(x_0, y_0)$.



(i) Show that the tangent at $P(x_1, y_1)$ has equation $xy_1 + yx_1 = 2$

2

(ii) Show that the chord of contact PQ has equation $xy_0 + yx_0 = 2$

2

(iii) Show that x_1 and x_2 are the roots of the quadratic equation $y_0 x^2 - 2x + x_0 = 0$

2

(iv) Hence, or otherwise, show that the midpoint R, of PQ has coordinates $\left(\frac{1}{y_0}, \frac{1}{x_0}\right)$

2

(v) Hence, or otherwise, show that as T moves on the hyperbola $xy = c^2$, $0 < c < 1$, R moves on the hyperbola $xy = \frac{1}{c^2}$

2

QUESTION 7 (15 marks) Start a new page

- (a) (i) Show that $\tan(A + \frac{\pi}{2}) = -\cot A$ 1
(ii) Use mathematical induction to prove that 4
 $\tan\left[(2n+1)\frac{\pi}{4}\right] = (-1)^n$ for all integer $n \geq 1$.
- (b) Two stones are thrown simultaneously from the same point in the same direction with the same non-zero angle of projection (upward inclination to the horizontal), α , but with different velocities U, V metres per second ($U < V$).
- The slower stone hits the ground at a point P on the same level as the point of projection. At that instant the faster stone just clears a wall of height h metres above the level of projection and its (downward) path makes an angle β with the horizontal.
- (i) Show that while the stones are in flight, the line joining them has a gradient of $\tan \alpha$. 3
(ii) Hence, express the horizontal distance from P to the foot of the wall in terms of h and α . 2
(iii) Show that $V(\tan \alpha + \tan \beta) = 2U \tan \alpha$. 3
(iv) Hence, deduce that, if $\beta = \frac{1}{2}\alpha$, then $U < \frac{3}{4}V$. 2

QUESTION 8 (15 marks) Start a new page

- (a) Show that the locus in the Argand plane represented by the equation $|z-1| + |z+1| = 4$ is a conic and find its cartesian equation. 3
- (b) A particle of mass m is projected against a constant gravitational force mg and resistance $\frac{mv}{k}$, where v is the velocity of the particle and k is a constant. Let x be the distance traveled in time t . Initially the particle has zero displacement and $v_0 = k(h - g)$, where h is a constant.
- (i) Show that the equation of motion of the particle is $\ddot{x} = -\left[\frac{kg+v}{k}\right]$ 1
(ii) Show that $t = k \log\left(\frac{kh}{kg+v}\right)$ 2
(iii) Find the time taken by the particle to reach the maximum height, H , and determine the height of that point. 3
- (c) A polynomial $P(x)$ is divided by $x^2 - a^2$ where $a \neq 0$, and the remainder is $px + q$.
- (i) Show that $p = \frac{1}{2a}[P(a) - P(-a)]$ 3
and $q = \frac{1}{2}[P(a) + P(-a)]$
- (ii) Find the remainder when $P(x) = x^n - a^n$, for n a positive integer, is divided by $x^2 - a^2$. 3

11(a) Let $u = 16 - x^2$

$\frac{du}{dx} = -2x$ $dx = -\frac{du}{2x}$

$\int \frac{x}{\sqrt{16-x^2}} dx = \int \frac{x}{\sqrt{u}} \cdot \frac{du}{-2x}$

$= -\frac{1}{2} \int u^{-\frac{1}{2}} du$

$= -\frac{1}{2} (2\sqrt{u}) + C$

$= -\sqrt{16-x^2} + C$

(b) $\int \frac{8}{x^2+4x+13} dx$

$= \int \frac{8}{(x+2)^2+9} dx$

$= \frac{8}{3} \tan^{-1}\left(\frac{x+2}{3}\right) + C$

(c) $\int x^4 \ln x dx$

$= \left[\frac{x^5}{5} \ln x \right] - \int \frac{x^5}{5} \cdot \frac{1}{x} dx$

$= \frac{e^5}{5} - \frac{1}{5} \left[\frac{x^5}{5} \right]_1^e$

$= \frac{e^5}{5} - \frac{e^5}{25} + \frac{1}{25} = \frac{4e^5+1}{25}$

(d) Let $u = \cos x$

$\frac{du}{dx} = -\sin x$ $du = -\sin x dx$

$\int \cos^2 x \sin^5 x dx$

$= \int \cos^2 x \sin^4 x \frac{du}{-1} (\sin x dx)$

$= \int u^2 (1-u^2)^2 - du$

$= -\int u^2 - 2u^4 + u^6 du$

Q2. (a) $z = 2+i$, $w = 1-i$

(i) $3z+iw$
 $= 3(2+i) + i(1-i)$
 $= 6+3i+i+i$
 $= 7+4i$

(ii) $z\bar{w}$
 $= (2+i)(1-i)$
 $= 2+i+2i-i$
 $= 1+3i$

(iii) $\frac{5}{z}$
 $= \frac{5}{2+i} \cdot \frac{2-i}{2-i}$
 $= \frac{10-5i}{4+i} = 2-i$

(b) $\alpha = -\sqrt{5} + i$

(i) $|\alpha| = (-\sqrt{5})^2 + 1^2$
 $= 6 = 2$

$\arg \alpha = \tan^{-1} \frac{1}{-\sqrt{5}}$
 $= \frac{5\pi}{6}$

$\alpha = 2 \operatorname{cis}\left(\frac{5\pi}{6}\right)$

(ii) $\alpha^4 = 2^4 \operatorname{cis}\left(4 \times \frac{5\pi}{6}\right)$
 $= 16 \operatorname{cis}\left(-\frac{2\pi}{3}\right)$

(iii) $\alpha^4 = 16\left(\cos -\frac{2\pi}{3} + i \sin -\frac{2\pi}{3}\right)$
 $= 16x - \frac{1}{2} + i(16x - \frac{\sqrt{3}}{2})$
 $= -8 - 8\sqrt{3}i$

(c) I

①

②

③

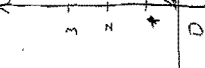
④

⑤

⑥

⑦

(c)



(d)(i) $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$a=4$, $b=3$

$b^2 = a^2(e^2 - 1)$

$9 = 16(e^2 - 1)$

$e^2 - 1 = \frac{9}{16}$

$e^2 = \frac{9}{16} + 1 = \frac{25}{16}$

$e = \frac{5}{4}$

$S \equiv (ae, 0)$, $S' \equiv (-ae, 0)$

$\equiv (5, 0)$ $\equiv (-5, 0)$

When $y=0$, $\frac{x^2}{16} = 1$

$x = \pm 4$

$\therefore x$ -intercepts are 4 & -4

(ii) Directions are

$x = \pm \frac{a}{e}$
 $= \pm \frac{16}{5}$

Eg. of asymptotes are

$y = \pm \frac{b}{a}x$
 $= \pm \frac{3}{4}x$

$x = 4 \sec \theta$
 $y = 3 \tan \theta$

(iii)

①

$$3(a) \quad x^3 + mx + n = 0.$$

$$\alpha\beta\gamma = -n$$

$$\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = m.$$

$$(i) \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{m}{-n}$$

$$(ii) \quad \begin{aligned} \alpha^3 + m\alpha + n &= 0 & \text{--- (1)} \\ \beta^3 + m\beta + n &= 0 & \text{--- (2)} \\ \gamma^3 + m\gamma + n &= 0 & \text{--- (3)} \end{aligned}$$

$$\begin{aligned} (1) + (2) + (3) \\ \alpha^3 + \beta^3 + \gamma^3 + m(\alpha + \beta + \gamma) + 3n &= 0 \\ \alpha^3 + \beta^3 + \gamma^3 &= -3n. \end{aligned}$$

$$(iii) \quad (\sqrt{x})^3 + m\sqrt{x} + n = 0$$

$$\sqrt{x}(x+m) + n = 0$$

$$\sqrt{x} = \frac{-n}{x+m}$$

$$x = \frac{n^2}{(x+m)^2}$$

$$x(x+m)^2 - n^2 = 0.$$

$$x^3 + 2mx^2 + m^2x - n^2 = 0.$$

$$(b) \quad P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$$

$$P'(x) = 4x^3 - 15x^2 - 18x + 81$$

$$P''(x) = 12x - 30x - 18 \quad (1)$$

$$\text{When } P''(x) = 0 \quad 12x - 30x - 18 = 0.$$

$$x = -\frac{1}{2} \text{ or } x = 3 \quad (1)$$

$$P'(-\frac{1}{2}) \neq P'(3) = 0.$$

$$\therefore x = 3 \text{ is a triple root of}$$

$$P(x) = 0 \quad (1)$$

$$\text{If the other root is } \alpha$$

$$\text{Sum of roots} = 9 + \alpha = 5.$$

$$\alpha = -4. \quad (1)$$

$$\therefore \text{Roots of } P(x) = 0 \text{ are } 3, 3, 3, + -4.$$

$$(c) \quad y = e^{-x} (A \sin 2x + B \cos 2x)$$

$$e^x y = A \sin 2x + B \cos 2x.$$

$$e^x \frac{dy}{dx} + y e^x = 2A \cos 2x - 2B \sin 2x.$$

$$e^x \frac{dy}{dx} + \frac{dy}{dx} e^x + y e^x = e^x \frac{dy}{dx} \quad (1)$$

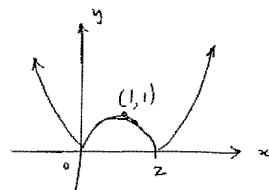
$$= -4A \sin 2x - 4B \cos 2x \quad (1)$$

$$= -4e^x y.$$

$$e^x \left(\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y \right) = 0.$$

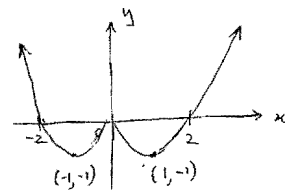
$$\therefore \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 5y = 0. \quad (1)$$

$$Q4. (a) (i) \quad y = |f(x)|$$

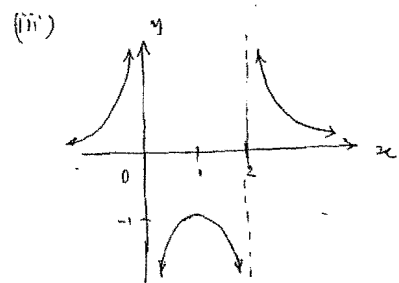


(1)

$$(ii) \quad y = f(|x|).$$

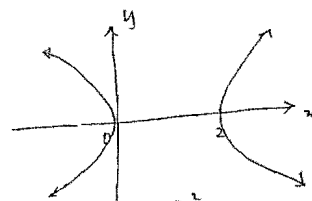


(1)



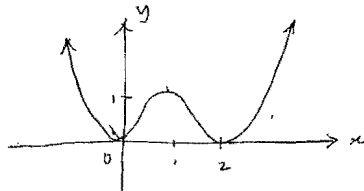
(2)

$$(iv) \quad y^2 = f(x).$$



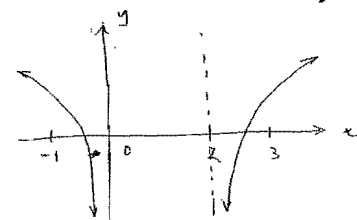
(2)

$$(v) \quad y = [f(x)]^2$$

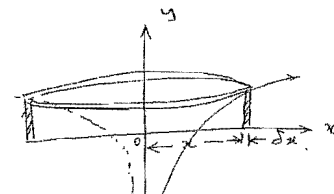


(1)

$$(vi) \quad y = \ln[f(x)].$$



(b)



$$\delta V = [\pi(x+\delta x)^2 - \pi x^2]$$

$$= 2\pi x y \delta x + \pi y \delta x^2$$

$$\approx 2\pi x y \delta x.$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=1}^r 2\pi x y \delta x$$

$$= \int_1^e 2\pi x y dx \quad (1)$$

$$= 2\pi \int_1^e x \ln x dx$$

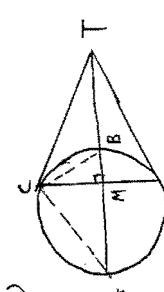
$$= 2\pi \left[\frac{x^2}{2} \ln x \right]_1^e - 2\pi \int_1^e \frac{x}{2} dx$$

$$= \pi \left[x^2 \ln x - \frac{x^2}{2} \right]_1^e$$

$$= \pi \left[(e^2 - \frac{e^2}{2}) - (0 - \frac{1}{2}) \right]$$

$$= \frac{\pi}{2} (e^2 + 1) \text{ unit}^3. \quad (2)$$

(c) $1+x+x^2+\dots+x^n = \frac{x^{n+1}-1}{x-1}$
 f. both sides w.r.t. x
 $2x+3x^2+\dots+n x^n = \frac{(n+1)x^{n+1}-(n+1)x^n-(x-1)}{(x-1)^2}$
 $= \frac{(n+1)x^{n+1}-(n+1)x^n-x+1}{(x-1)^2}$
 $= \frac{n x^{n+1}-(n+1)x^n+1}{(x-1)^2}$
 + $x=2$
 $2 \times 2 + 3 \times 4 + 4 \times 8 + \dots + n \times 2^n$
 $= \frac{n \times 2^{n+1} - (n+1)2^{n+1} + 1}{1}$
 $= n \times 2^{n+1} - 2^n(n+1) + 1$
 $= 2^n [2n - (n+1)] + 1$
 $= 2^n (n+1) + 1$



Join AC + BC
 $\angle ACB = 90^\circ$ ($\angle$ in a semi-c.)
 $\angle TCB = \angle BAC$ ($\angle$ bet. tangent + chord = α = $\angle$ in alt. seg.)
 $\angle ACM = 90^\circ - \alpha$ ($\angle$ sum of Δ)
 But $\angle ACB = 90^\circ$
 $\therefore \angle BCM = 90^\circ - (90^\circ - \alpha) = \alpha$
 $\therefore \angle TCB = \angle BCM$
 i.e. BC bisects $\angle TCM$.

(ii) In $\Delta BCM + \Delta CAM$
 $\angle BMC = \angle AMC = 90^\circ$ ($CM \perp AB$)
 $\angle CAM = \angle BCM = \alpha$ (proved above)
 $\therefore \Delta BCM \sim \Delta CAM$ (equiangular)
 $\frac{CM}{AM} = \frac{BM}{CM}$ (corresp sides of similar Δ s)
 $\therefore CM^2 = AM \times BM$
 (iii) $CT^2 = TA \times TB$
 (The sq. on the tangent equal the product of the segs. of any chord through the pt.)
 In ΔTCM
 $CT^2 = CM^2 + MT^2$
 $= AM \times BM + MT^2$
 $TA \times TB = CT^2$
 $= AM \times BM + MT^2$
 $= (TA - TM) \times MB + MT^2$
 $= TA \times MB - TM \times MB + MT^2$
 $= TA \times MB - TM(TM - MB)$
 $= TA \times MB - TM \times TB$
 $TA \times TB - TM \times TB = TA \times MB$
 $TB(TA - TM) = TA \times MB$
 $TB \times AM = TA \times MB$
 $\frac{AM}{MB} = \frac{TA}{TB}$
 i.e. M divides AB internally in the same ratio as T divides BR externally.
 (c) (i) $I_n = \int \tan^n dx$
 $= \int \tan^{n-2} (\sec^2 x - 1) dx$
 $= \int \tan^{n-2} \sec^2 x - \int \tan^{n-2} dx$
 $= \frac{\tan^{n-1}}{n-1} - I_{n-2}$
 (ii) $I_6 = \frac{1}{5} \tan^5 x - I_4$
 $= \frac{1}{5} \tan^5 x - \left[\frac{1}{3} \tan^3 x - I_2 \right]$
 $= \frac{1}{5} \tan^5 x - \frac{1}{3} \tan^3 x + \tan x - I_0$
 $= \frac{1}{5} \tan^5 x - \frac{1}{3} \tan^3 x + \tan x - x + C$

Q6 (a) $\delta V = \frac{1}{2} \pi y \delta x$
 $= \frac{1}{2} (4-x^2) \sqrt{4-x^2} \delta x$
 $= \frac{1}{2} (4-x^2)^{3/2} \delta x$
 $V = \lim_{\delta x \rightarrow 0} \sum_{x=-2}^2 \frac{1}{2} (4-x^2)^{3/2} \delta x$
 $= \int_{-2}^2 \frac{1}{2} (4-x^2)^{3/2} dx$
 $= \int_0^2 (4-x^2)^{3/2} dx$ [($4-x^2$)^{3/2} is an even function]
 Let $x = 2 \sin \theta$ $\frac{dx}{d\theta} = 2 \cos \theta$
 $4-x^2 = 4(1-\sin^2 \theta) = 4 \cos^2 \theta$
 $x=0, 2 \sin \theta = 0 \Rightarrow \theta = 0$
 $x=2, 2 \sin \theta = 2 \Rightarrow \theta = \frac{\pi}{2}$
 $V = \int_0^{\pi/2} (4 \cos^2 \theta)^{3/2} \cdot 2 \cos \theta d\theta$
 $= 16 \int_0^{\pi/2} \cos^4 \theta d\theta$
 $= 16 \int_0^{\pi/2} \frac{\cos 2\theta + 1}{2} d\theta$
 $= 4 \int_0^{\pi/2} (\cos 2\theta + 1) d\theta$
 $= 4 \left[\frac{\sin 2\theta}{2} + \theta \right]_0^{\pi/2}$
 $= 2 \left[\left(0 + \frac{3\pi}{2}\right) - (0 + 0) \right]$
 $= 3\pi \text{ unit}^2$

(b) (i) $xy=1 \Rightarrow \frac{dy}{dx} = -\frac{1}{x}$
 At (x_1, y_1) $m_T = -\frac{1}{x_1}$
 Eqn of tangent at (x_1, y_1) is
 $y - y_1 = -\frac{1}{x_1} (x - x_1)$
 But $x_1 y_1 = 1 \Rightarrow y_1 = \frac{1}{x_1}$
 $\therefore x_1 y - 1 = -x y + 1$
 $x_1 y + x y = 2$
 (ii) Tangents at P & Q are
 $x_1 y_0 + x_0 y = 2$
 $x_2 y_0 + x_0 y = 2$
 Eqn $x y_0 + x_0 y = 2$ satisfied by both tangents P & Q.
 $\therefore x y_0 + x_0 y = 2$ is eqn of PQ.
 (iii) x_1, x_2 are solution to the set of simultaneous eqns
 $x y_0 + x_0 y = 2$ — (1)
 and $xy = 1$ — (2)
 Sub $y = \frac{1}{x}$ into (1)
 $x y_0 + \frac{x_0}{x} = 2$
 $x^2 y_0 + x_0 = 2x$
 $x^2 y_0 - 2x + x_0 = 0$
 (iv) $R = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$
 from (iii) $x_1 + x_2 = \frac{2}{y_0}$ (sum of roots)
 $\therefore \frac{x_1+x_2}{2} = \frac{1}{y_0}$
 Sub into eqn of PQ
 $\frac{1}{y_0} \cdot y_0 + x_0 y = 2$
 $1 + x_0 y = 2$
 $y = \frac{1}{x_0}$
 $\therefore R = \left(\frac{1}{x_0}, \frac{1}{y_0} \right)$

Q6(c)(v) If T moves on $xy = c^2$.

then $x_0 y_0 = c^2$.

For R $x = \frac{1}{y_0}$, $y = \frac{1}{x_0}$.

$\therefore R$ moves on $xy = \frac{1}{c^2}$ which is a hyperbola.

$$\begin{aligned} 2T(2V \tan(A + \frac{\pi}{2})) &= -\tan(\pi - (A + \frac{\pi}{2})) \\ &= -\tan(\frac{\pi}{2} - A) = -\cot A \end{aligned}$$

$$\text{(ii)} \quad \tan \left[(2n+1)\frac{\pi}{4} \right] = (-1)^n$$

$$\text{For } n=1 \quad \text{LHS} = \tan \frac{3\pi}{4} = -1$$

$$\text{RHS} = (-1)^1 = -1 = \text{LHS}$$

$\therefore$ True for $n=1$.

Assume true for $n=k$.

$$\tan \left[(2k+1)\frac{\pi}{4} \right] = (-1)^k$$

For $n=k+1$, need to prove

$$\tan \left[(2k+3)\frac{\pi}{4} \right] = (-1)^{k+1}$$

$$\text{HS} = \tan \left[(2k+3)\frac{\pi}{4} \right]$$

$$= \tan \left[(2k+1)\frac{\pi}{4} + \frac{\pi}{2} \right]$$

$$= -\cot \left[(2k+1)\frac{\pi}{4} \right]$$

$$= -\frac{1}{\tan \left[(2k+1)\frac{\pi}{4} \right]} = -\frac{1}{(-1)^k} = (-1)^{k+1}$$

$$= (-1)^{k+1} = \text{RHS}$$

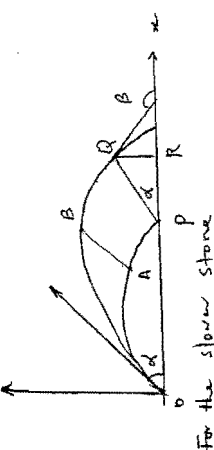
$\therefore$ True for $n=k+1$ if true for $n=k$.

$\therefore$ True for $n=1$, hence true for

$n=2, 3, 4, \dots$

$\therefore$ True for $n \geq 1$.

(b)(ii)



For the slower stone
 $y_1 = -gt^2 + ut \sin \alpha$
 $y_1 = -\frac{1}{2}gt^2 + ut \sin \alpha$

Similarly for the faster stone
 $y_2 = -\frac{1}{2}gt^2 + vt \sin \alpha$, $x_2 = vt \cos \alpha$

At time t , let the stones be at A & B.

$$M_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-\frac{1}{2}gt^2 + vt \sin \alpha - (-\frac{1}{2}gt^2 + ut \sin \alpha)}{vt \cos \alpha - ut \cos \alpha}$$

$$= \frac{t \sin \alpha (v - u)}{t \cos \alpha (v - u)} = \tan \alpha$$

(iii) When the faster stone is at Q, the top of the wall, the slower stone is at P.

$$m_{PQ} = \tan \alpha, \therefore \angle QPR = \alpha$$

$$\tan \alpha = \frac{QR}{PR} = \frac{h}{PR}$$

$$\therefore PR = \frac{h}{\tan \alpha} = h \cot \alpha$$

(iii) When $y_1 = 0$.

$$-\frac{1}{2}gt^2 + ut \sin \alpha = 0$$

$$gt^2 = 2ut \sin \alpha$$

$$t = \frac{2u \sin \alpha}{g} \quad (t \neq 0)$$

At this instant, the faster stone is at Q.

$$\begin{aligned} \tan(\pi - \beta) &= -\frac{y}{x} \\ \therefore \tan \beta &= -\frac{y}{x} \\ \tan \beta &= \frac{-(-gt + V \sin \alpha)}{V \cos \alpha} \\ &= \frac{g(2u \sin \alpha)}{V \cos \alpha} - V \sin \alpha \\ &= \frac{g \sin \alpha (2u - V)}{V \cos \alpha} = \tan \alpha (2u - V) \end{aligned}$$

$$\begin{aligned} V \tan \beta &= \tan \alpha (2u - V) \\ \therefore (\tan \alpha + \tan \beta) &= 2u \tan \alpha \\ \text{(ii) If } \beta &= \frac{1}{2} \alpha \\ \tan \alpha &= \frac{2 \tan \beta}{1 - \tan^2 \beta} \\ V \left(\frac{2 \tan \beta}{1 - \tan^2 \beta} + \tan \beta \right) &= 2u \left(\frac{2 \tan \beta}{1 - \tan^2 \beta} \right) \\ 2V \tan \beta + V \tan \beta (1 - \tan^2 \beta) &= 4u \tan \beta \\ 2V + V(1 - \tan^2 \beta) &= 4u \\ u &= \frac{1}{4}V(3 - \tan^2 \beta) \\ &< \frac{3}{4}V \quad (\because \tan^2 \beta > 0) \end{aligned}$$

Q8(a) $P = z$, $S = 1$, $S' = -1$.
 $PS = |z - 1|$, $PS' = |z + 1|$
If $|z - 1| + |z + 1| = 4$, then
 $PS + PS' = 4$.
The locus of P is therefore an ellipse with foci S, S' and major axis 4 units.
i.e. $a = 2$.
 $ae = 1$ $\therefore e = \frac{1}{2}$.
 $b^2 = a^2(1 - e^2) = 3$.
 $\therefore$ Cartesian eqn of P is

$$\begin{aligned} \text{(b)} \quad \uparrow m\ddot{x} \quad v_0 &= k(h - g) \\ \downarrow \frac{m\dot{v}}{k} \quad \text{(i)} \quad m\ddot{x} &= -mg - \frac{m\dot{v}}{k} \\ \ddot{x} &= -\left(\frac{kg + \dot{v}}{k}\right) \quad \text{①} \\ \text{(ii)} \quad \frac{dv}{dt} &= -\left(\frac{kg + v}{k}\right) \\ \frac{dt}{dv} &= \frac{-k}{kg + v} \\ t &= -k \int_v^v \frac{1}{kg + v} dv \\ &= -k \left[\ln(kg + v) \right]_v^v \quad \text{①} \\ &= -k \left[\ln(kg + v) - \ln(kg + v_0) \right] \\ &= -k \ln \left(\frac{kg + v}{kg + v_0} \right) \\ &= k \ln \left(\frac{kg + v_0}{kg + v} \right) \\ &= k \ln \left(\frac{kg + h - g}{kg + v} \right) \quad \text{②} \\ \text{(iii) Max. height when } v &= 0 \\ t &= \ln \left(\frac{kg + h - g}{kg} \right) = k \ln \left(\frac{h}{g} \right) \quad \text{③} \\ v \frac{dv}{dx} &= -\left(\frac{kg + v}{k}\right) \\ \frac{dx}{dv} &= \frac{-kv}{kg + v} \\ x &= -k \int \frac{v}{kg + v} dv \\ &= -k \int \left(1 - \frac{kg}{kg + v} \right) dv \\ &= -k \left[v - kg \ln(kg + v) \right] + c \quad \text{④} \\ \text{When } x=0, v &= kh - g \\ 0 &= -k \left[(kh - g) - kg \ln(kg + kh - g) \right] + c \\ c &= (kh - g - kg \ln(kg + kh - g))k \end{aligned}$$

$$\begin{aligned} \text{Q8(b)(iii) Cont.} \\ x &= -k \left[v - kg \ln(kg + v) \right] + k \left[kh - g - kg \ln(kg + kh - g) \right] \\ &= k \left[kh - g - v + kg \ln \left(\frac{kg + v}{kg + kh - g} \right) \right] \\ \text{At } x &= H, v = 0 \\ H &= k \left[kh - g + kg \ln \left(\frac{kg}{kg + kh - g} \right) \right] \quad \text{①} \\ \text{(c)(i)} \quad P(x) &= (x^2 - a^2)Q(x) + px + q \\ P(a) &= pa + q \quad \text{①} \\ P(-a) &= -pa + q \quad \text{②} \\ \text{①} - \text{②} \\ P(a) - P(-a) &= 2pa \\ \therefore p &= \frac{1}{2a} [P(a) - P(-a)] \quad \text{③} \\ \text{①} + \text{②} \\ P(a) + P(-a) &= 2q \\ \therefore q &= \frac{1}{2} [P(a) + P(-a)] \quad \text{④} \\ \text{(ii)} \quad x^n - a^n &= (x^2 - a^2)Q(x) + px + q \\ P(x) &= x^n - a^n \\ \text{When } n \text{ is even } P(a) &= 0 \\ P(-a) &= 0 \\ \therefore q &= 0, p = 0. \\ \therefore \text{no remainder.} \quad \text{①} \\ \text{When } n \text{ is odd} \\ P(a) &= 0, P(-a) = -2a^n \quad \text{②} \\ px + q &= \frac{1}{2a} [2a^n]x + \frac{1}{2}(0 - 2a^n) \\ &= a^{n-1}x - a^n \quad \text{③} \end{aligned}$$